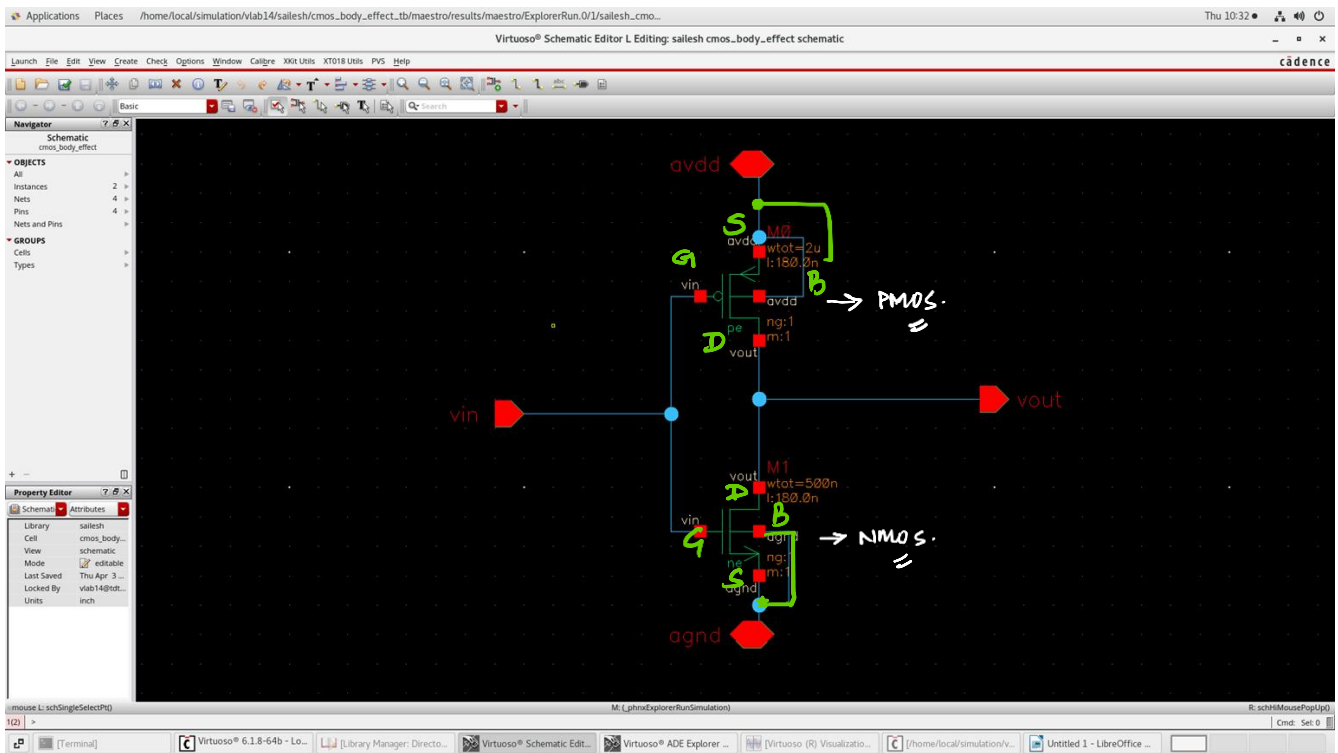
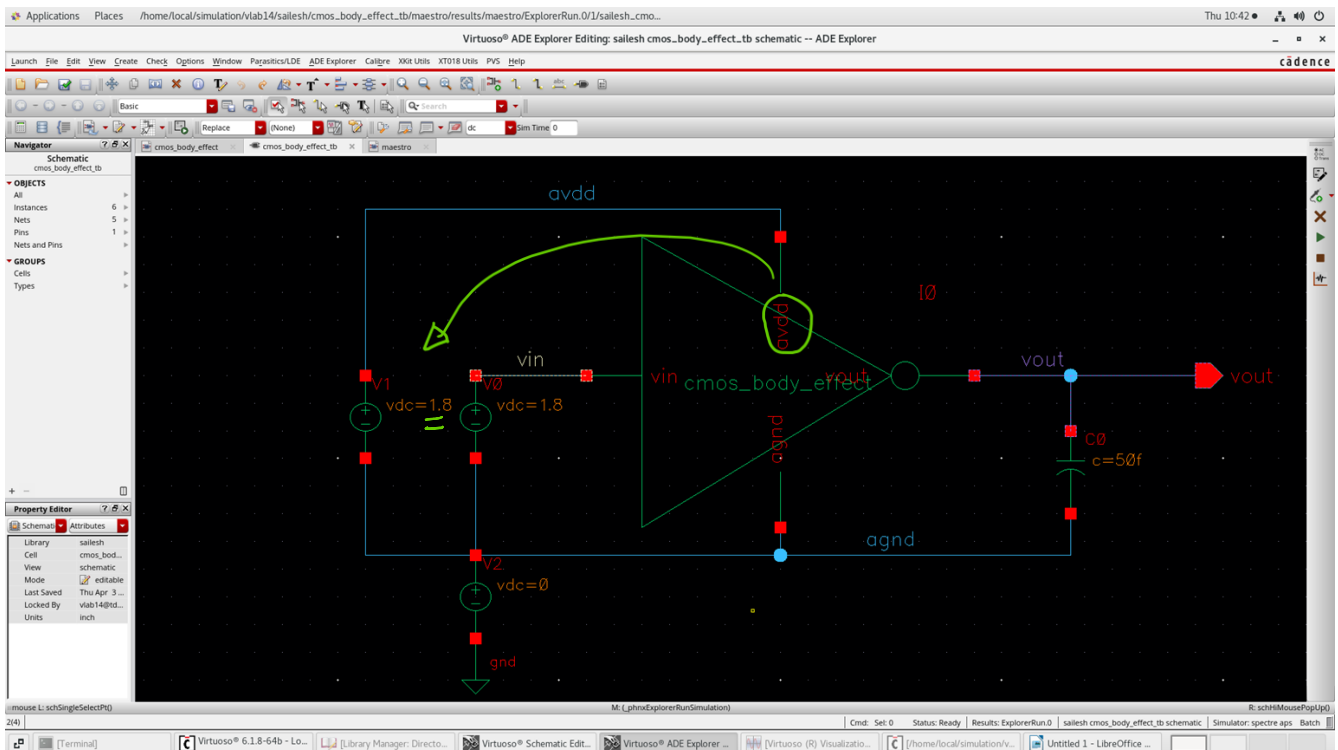
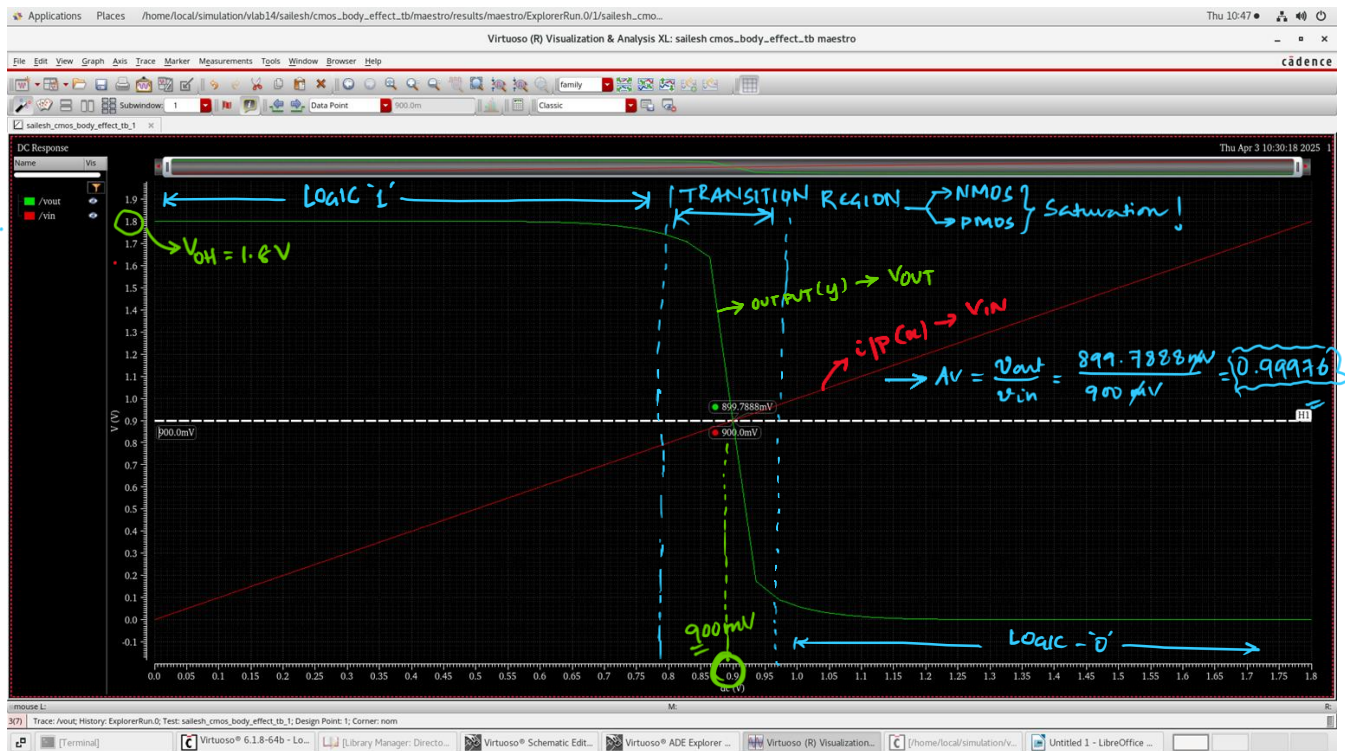


CMOS INVERTER BODY EFFECT DEMONSTRATION & it's RELATION WITH PREFERENCE of WELL FORMATION - DEEP WELL CMOS FABRICATION



$$W_p/L_p = 2\mu/180n : W_n/L_n = 500n/180n$$

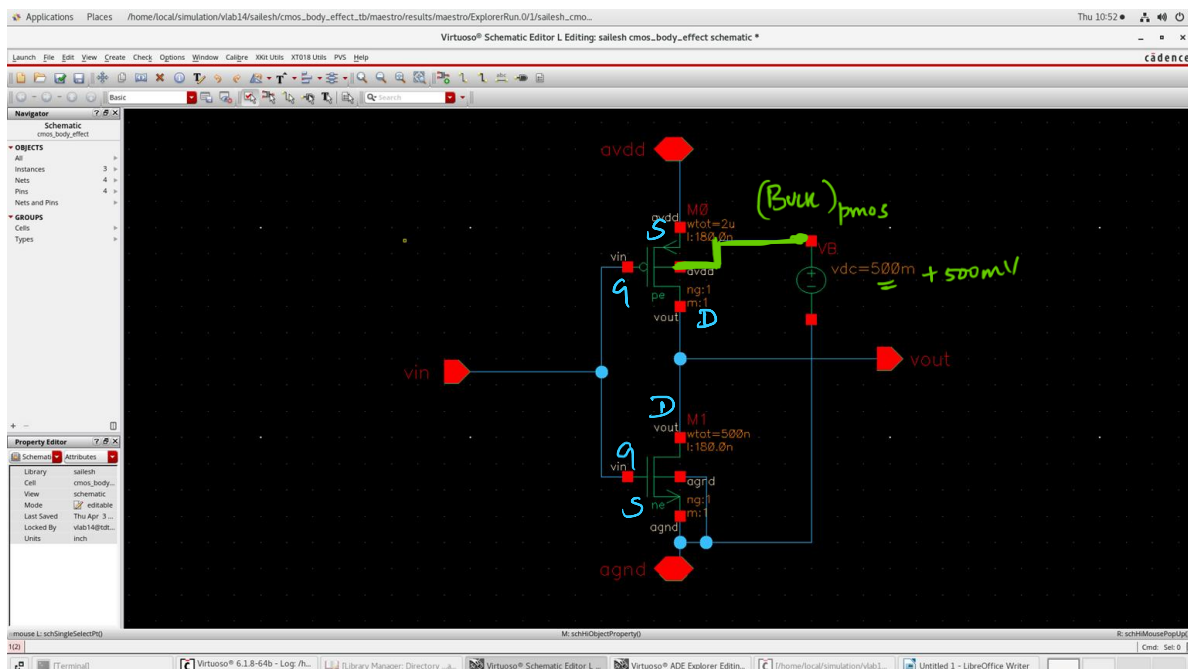


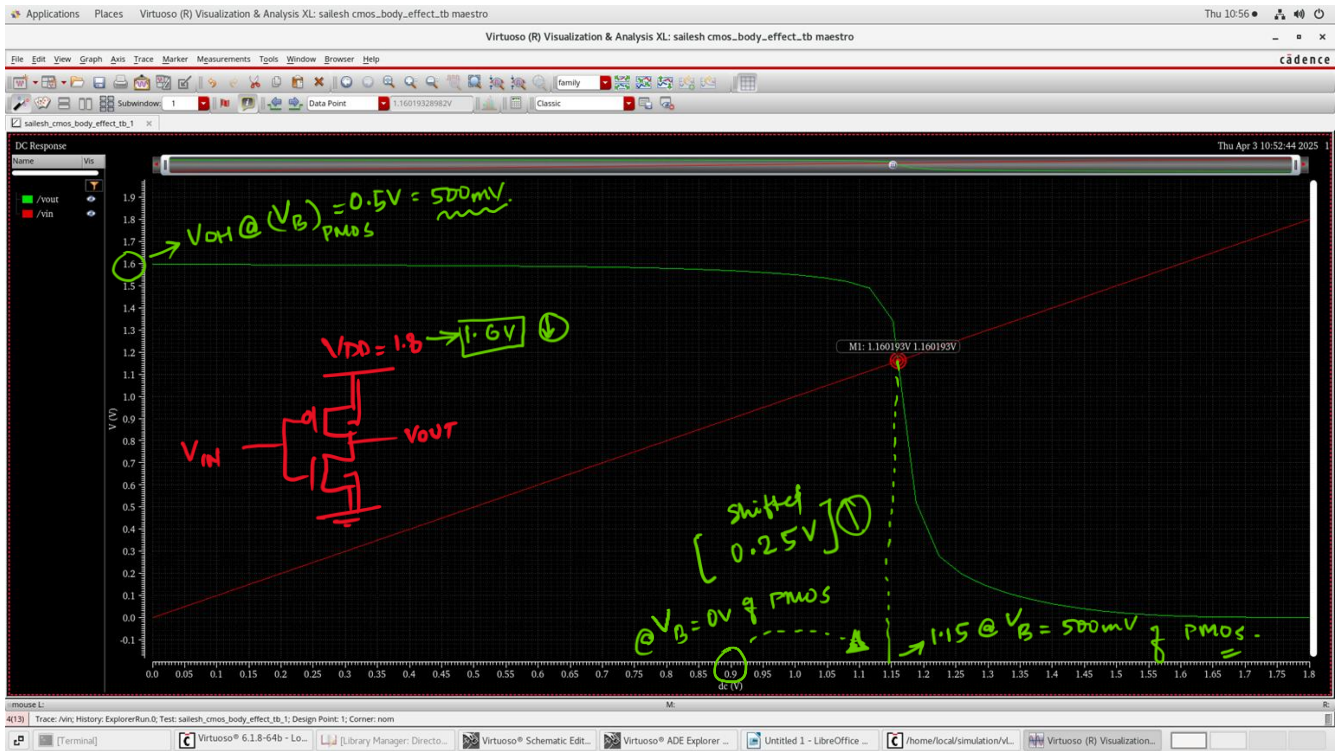


$$A_v = v_{out}/v_{in} = 899.7888/900mV = 0.99976$$

Now, Let's give some body bias to PMOS & NMOS case wise, First off all to PMOS, Let's apply 500mV to VB then, 500mV to NMOS.

1. CASE 01 : PMOS VB = +500mV

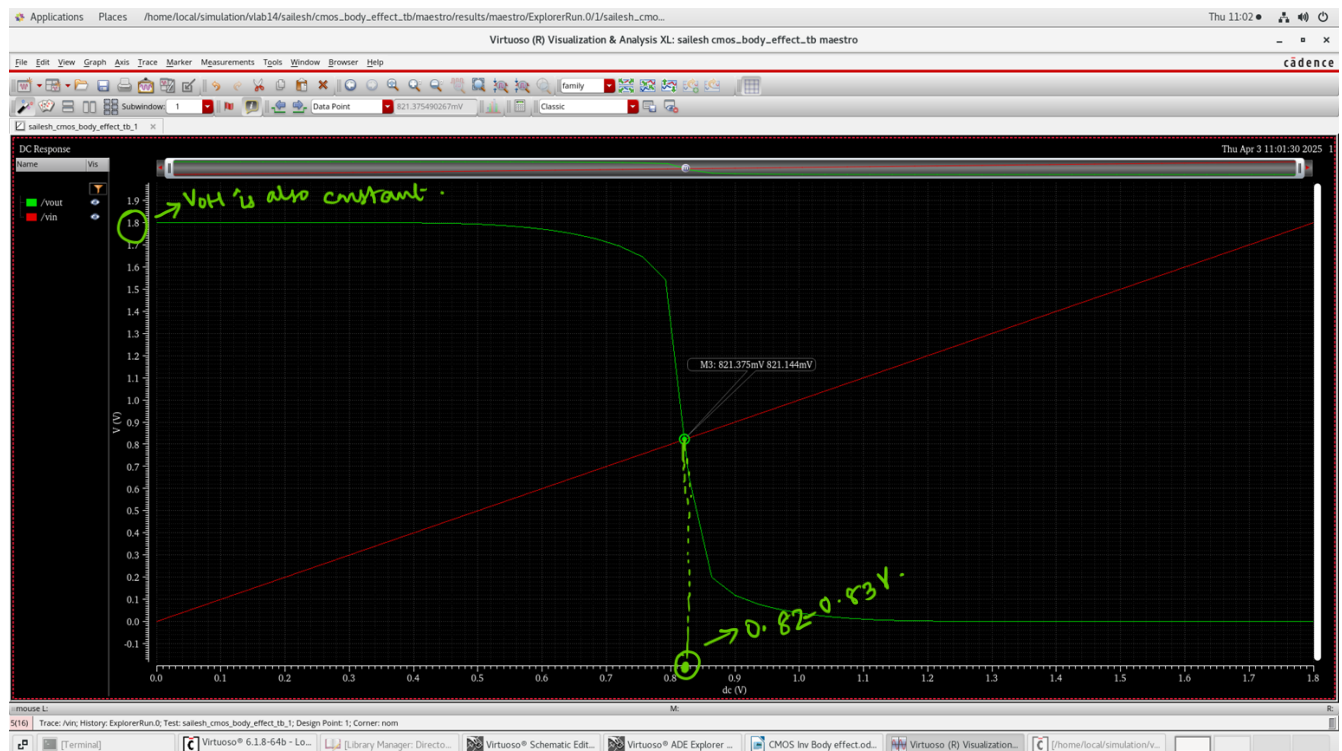
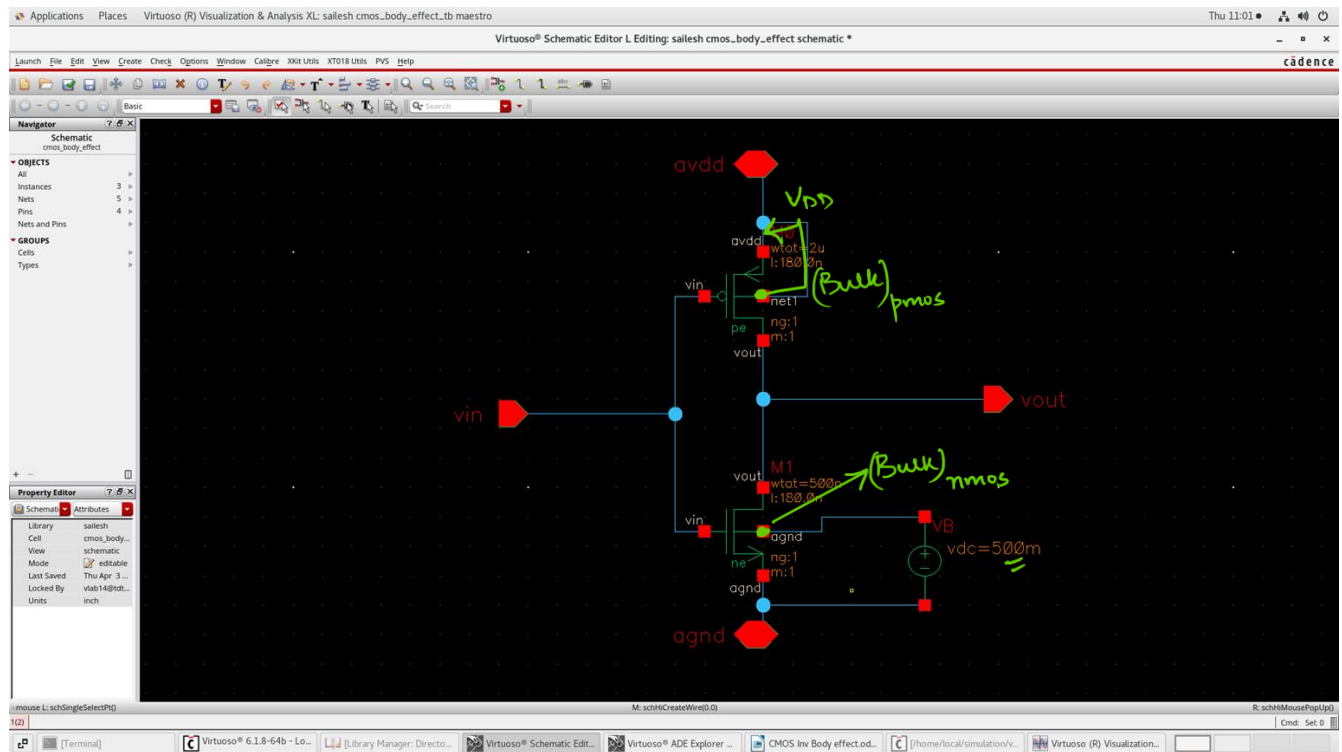




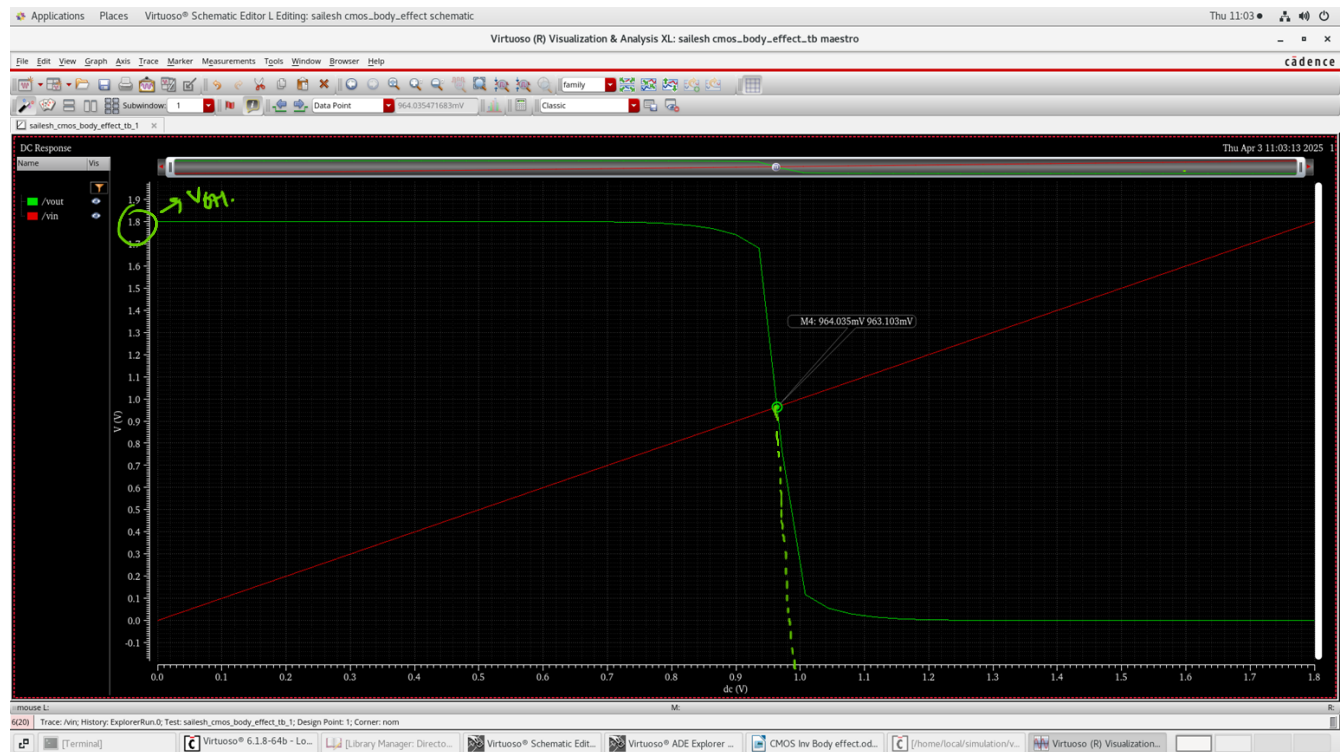
2. CASE 02 : PMOS $V_B = -500mV$



3. CASE 03 : NMOS VB = +500mV



4. CASE 04 : NMOS VB = -500mV



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